Experiment 12

```python
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# ----------------------------------------
# 1. Create the original array
# ----------------------------------------

array = np.array([84, 2, 54, 90, 24, 16, 43, 8, 60, 69, 23, 65, 51, 47, 49, 15])

print("Original Array:")
print(array)

# ----------------------------------------
# 2. Descriptive statistics
# ----------------------------------------

print("\nMean:")
print(np.mean(array))

q1 = np.percentile(array, 25)
q2 = np.percentile(array, 50)
q3 = np.percentile(array, 75)
q4 = np.percentile(array, 100)

print("\n25th Percentile (Q1):")
print(q1)

print("\n50th Percentile (Median):")
print(q2)

print("\n75th Percentile (Q3):")
print(q3)

print("\n100th Percentile (Maximum):")
print(q4)

# ----------------------------------------
# 3. Outlier boundaries using the IQR
# ----------------------------------------

iqr = q3 - q1
lr = q1 - 1.5 * iqr
ur = q3 + 1.5 * iqr

print("\nOriginal Array Outlier Boundaries:")
print("Lower Range:", lr)
print("Upper Range:", ur)

# ----------------------------------------
# 4. Plot the original distribution
# ----------------------------------------

sns.histplot(array, kde=True)
plt.title("Distribution of Original Array")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```

```
# ----------------------------------------
# 5. Remove the outliers
# ----------------------------------------

new_array = array[(array >= lr) & (array <= ur)]

print("\nArray After Removing Outliers:")
print(new_array)

# ----------------------------------------
# 6. Plot the distribution after removing outliers
# ----------------------------------------

sns.histplot(new_array, kde=True)
plt.title("Distribution After Removing Outliers")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()

# ----------------------------------------
# 7. Recalculate the outlier boundaries
# ----------------------------------------

q1 = np.percentile(new_array, 25)
q3 = np.percentile(new_array, 75)
iqr = q3 - q1
lr1 = q1 - 1.5 * iqr
ur1 = q3 + 1.5 * iqr

print("\nNew Outlier Boundaries:")
print("Lower Range:", lr1)
print("Upper Range:", ur1)

# ----------------------------------------
# 8. Filter again to get the final array
# ----------------------------------------

final_array = new_array[
    (new_array >= lr1) & (new_array <= ur1)
]

print("\nFinal Array:")
print(final_array)


# ----------------------------------------
# 9. Plot the final distribution
# ----------------------------------------

sns.histplot(final_array, kde=True)
plt.title("Final Array Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```

✓ 2.1s

```
Original Array:
[84  2 54 90 24 16 43  8 60 69 23 65 51 47 49 15]

Mean:
43.75
```

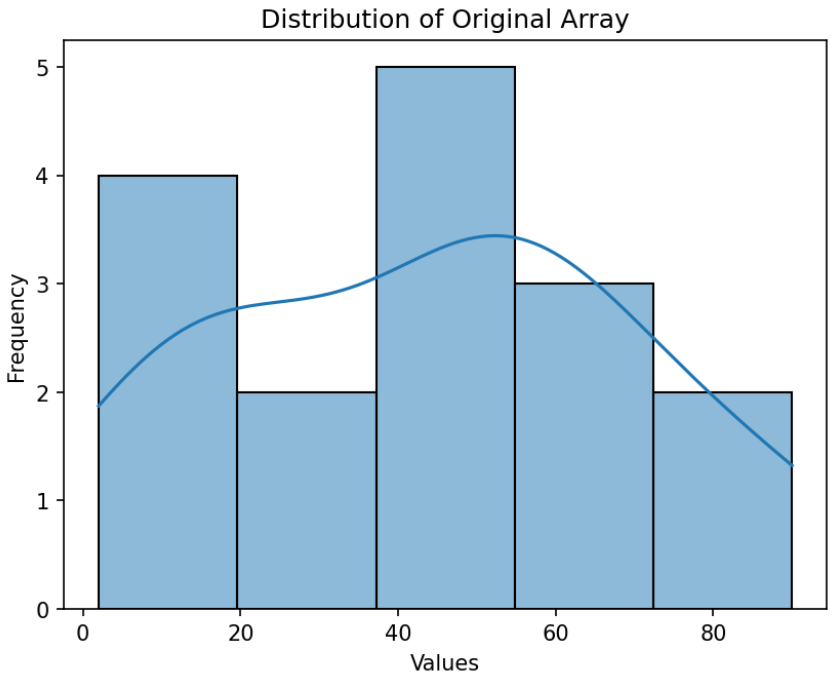
```
25th Percentile (Q1):
21.25

50th Percentile (Median):
48.0

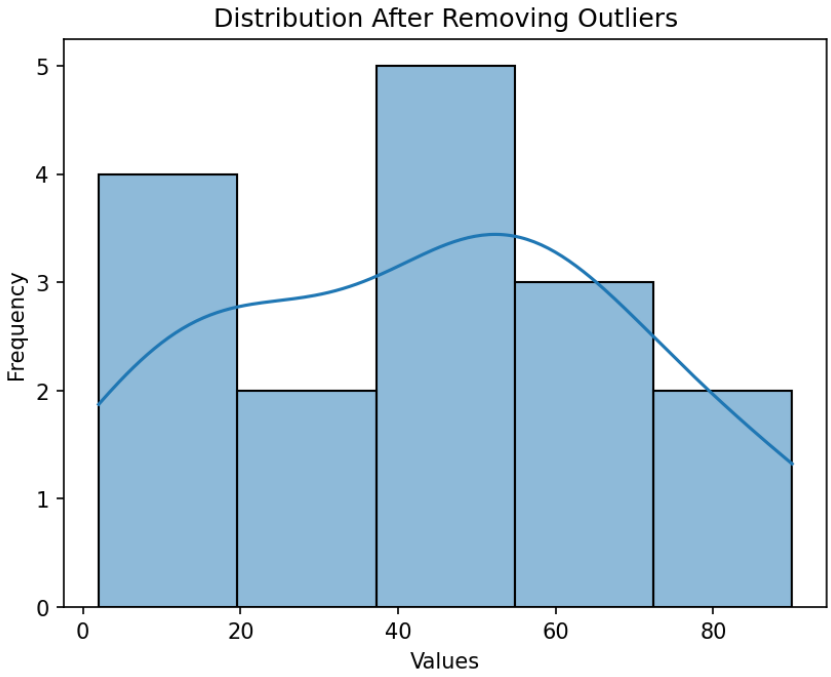
75th Percentile (Q3):
61.25

100th Percentile (Maximum):
90.0

Original Array Outlier Boundaries:
Lower Range: -38.75
Upper Range: 121.25
```




```
Array After Removing Outliers:
[84  2 54 90 24 16 43  8 60 69 23 65 51 47 49 15]
```




```
New Outlier Boundaries:
Lower Range: -38.75
Upper Range: 121.25

Final Array:
[84  2 54 90 24 16 43  8 60 69 23 65 51 47 49 15]
```

